

Interlude: Maximum Likelihood Estimation & the EM-algorithm

Maximum Likelihood & EM slides taken from UW CSE312 (winter '17)

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Parameter Estimation

Given: independent samples $x_1, x_2, \dots, x_n$ from a parametric distribution $f(x|\theta)$

Not formally “conditional probability,”
but the notation is convenient...

Goal: estimate θ .

E.g.: Given sample HHTTTTHTHTTTTHH of (possibly biased) coin flips, estimate

θ = probability of Heads

$f(x|\theta)$ is the Bernoulli probability mass function with parameter θ

Likelihood

(For *Discrete* Distributions)

$P(x \mid \theta)$: Probability of event x given *model* θ

Viewed as a function of x (fixed θ), it's a *probability*

$$\text{E.g., } \sum_x P(x \mid \theta) = 1$$

Viewed as a function of θ (fixed x), it's called *likelihood*

E.g., $\sum_{\theta} P(x \mid \theta)$ can be anything; *relative* values are the focus.

E.g., if θ = prob of heads in a sequence of coin flips then

$$P(\text{HHTHH} \mid .6) > P(\text{HHTHH} \mid .5),$$

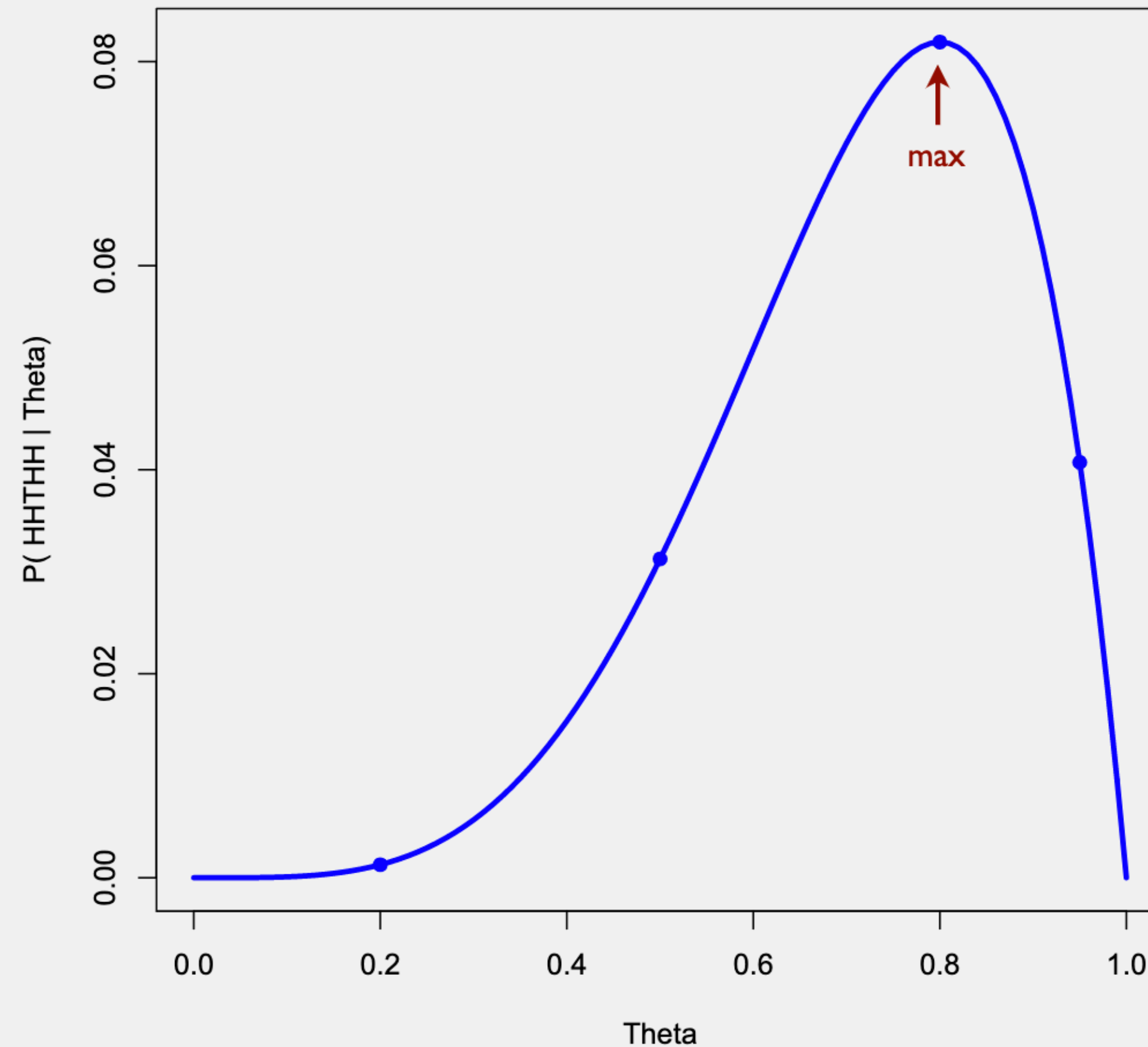
I.e., event HHTHH is *more likely* when $\theta = .6$ than $\theta = .5$

And **what θ make HHTHH *most* likely?**

Likelihood Function

$P(\text{HHTHH} \mid \theta)$:
Probability of HHTHH,
given $P(H) = \theta$:

θ	$\theta^4(1-\theta)$
0.2	0.0013
0.5	0.0313
0.8	0.0819
0.95	0.0407



Maximum Likelihood Parameter Estimation

(For *Discrete* Distributions)

One (of many) approaches to param. est.

Likelihood of (indp) observations $x_1, x_2, \dots, x_n$

$$L(x_1, x_2, \dots, x_n \mid \theta) = \prod_{i=1}^n f(x_i \mid \theta) \quad (*)$$

As a function of θ , what θ maximizes the likelihood of the data actually observed?

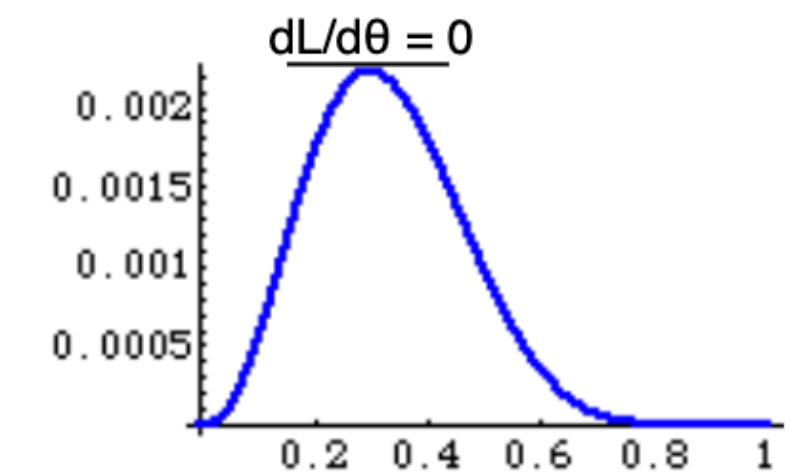
Typical approach: $\frac{\partial}{\partial \theta} L(\vec{x} \mid \theta) = 0$ or $\frac{\partial}{\partial \theta} \log L(\vec{x} \mid \theta) = 0$

(*) In general, (discrete) likelihood is the *joint* pmf; product form follows from independence

Example 1

n independent coin flips, $x_1, x_2, \dots, x_n$; n_0 tails, n_1 heads,
 $n_0 + n_1 = n$; θ = probability of heads

$$L(x_1, x_2, \dots, x_n \mid \theta) = (1 - \theta)^{n_0} \theta^{n_1}$$



$$\log L(x_1, x_2, \dots, x_n \mid \theta) = n_0 \log(1 - \theta) + n_1 \log \theta$$

$$\frac{\partial}{\partial \theta} \log L(x_1, x_2, \dots, x_n \mid \theta) = \frac{-n_0}{1 - \theta} + \frac{n_1}{\theta}$$

Setting to zero and solving:

$$\hat{\theta} = \frac{n_1}{n}$$

Observed fraction of
successes in sample is
MLE of success
probability in population

(Also verify it's max, not min, & not better on boundary)

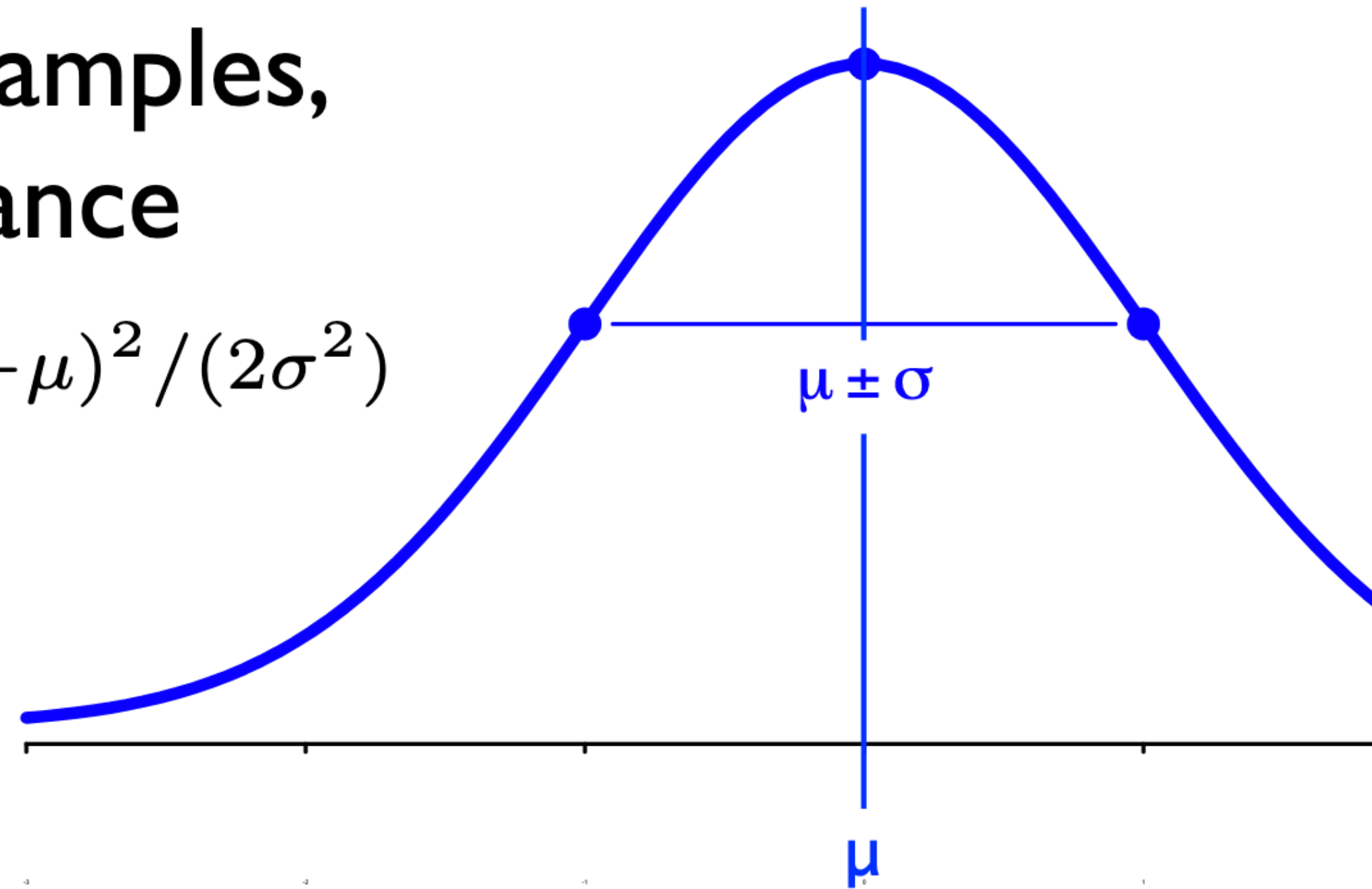
Parameter Estimation

Given: indp samples $x_1, x_2, \dots, x_n$ from a parametric distribution $f(x|\theta)$, **estimate:** θ .

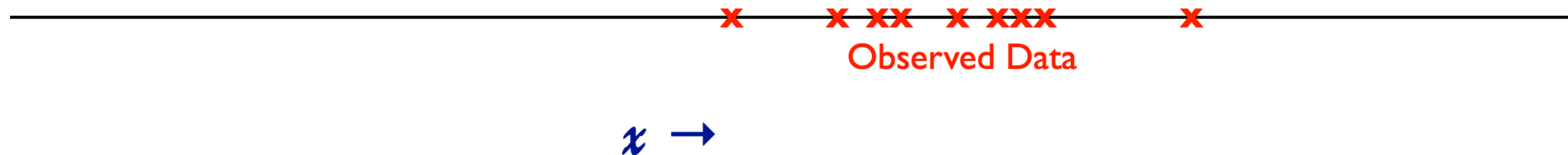
E.g.: Given n *normal* samples, estimate mean & variance

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2 / (2\sigma^2)}$$

$$\theta = (\mu, \sigma^2)$$

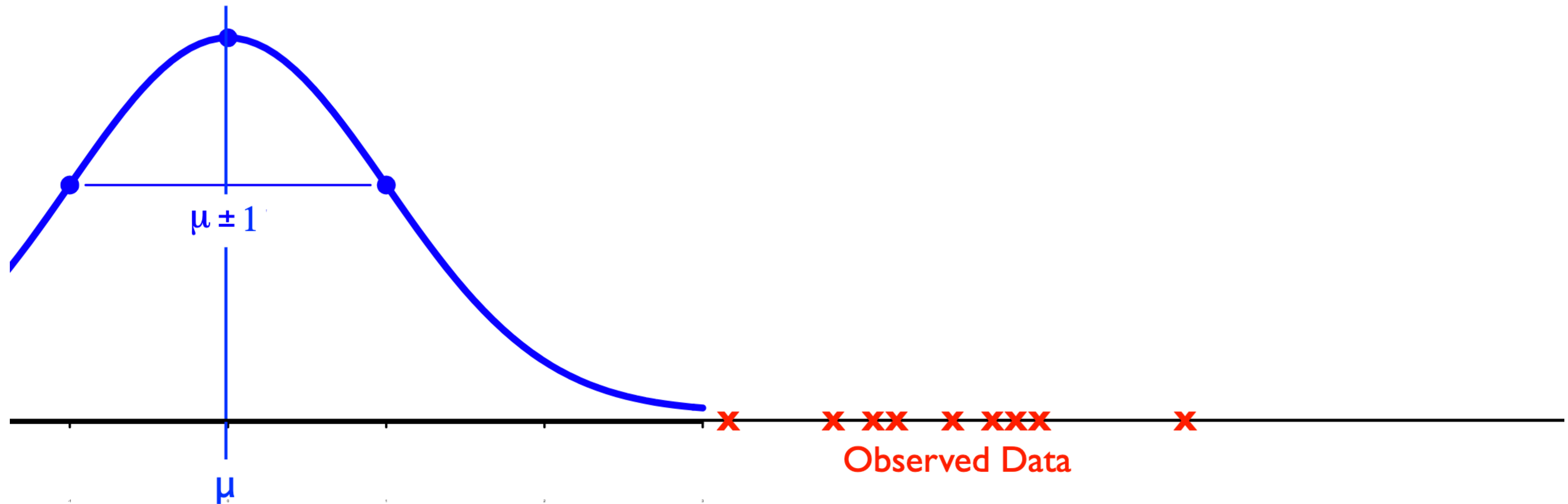


Ex2: I got data; a little birdie tells me
it's normal, and promises $\sigma^2 = 1$



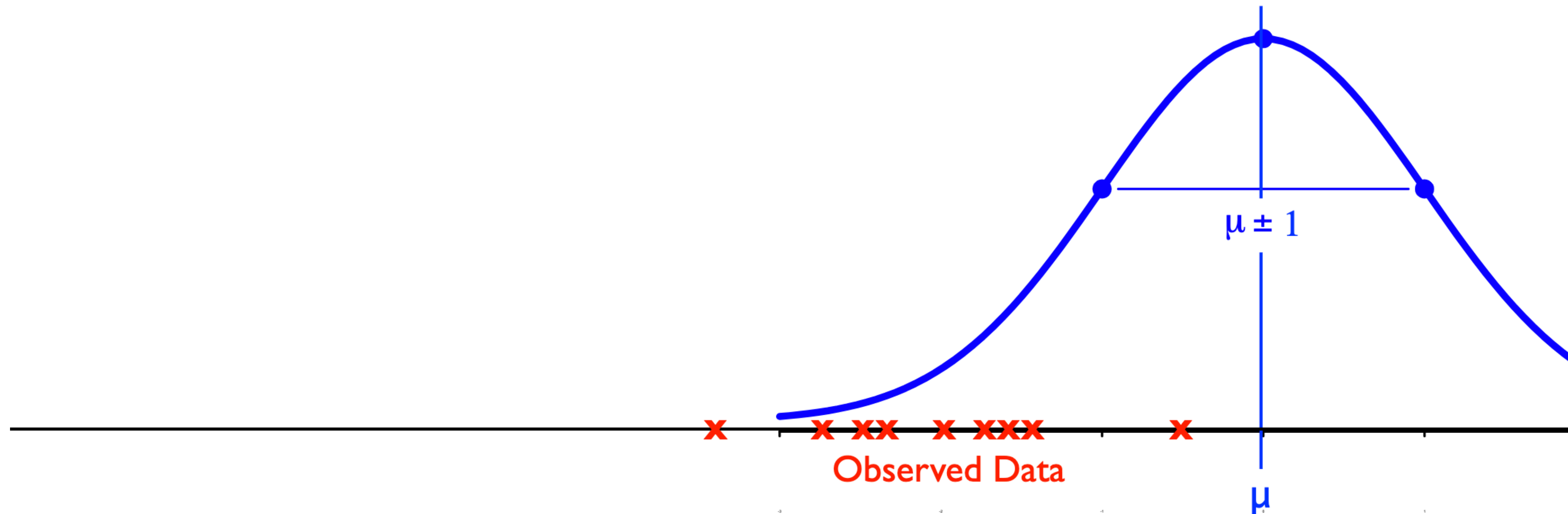
Which is more likely: (a) this?

μ unknown, $\sigma^2 = 1$



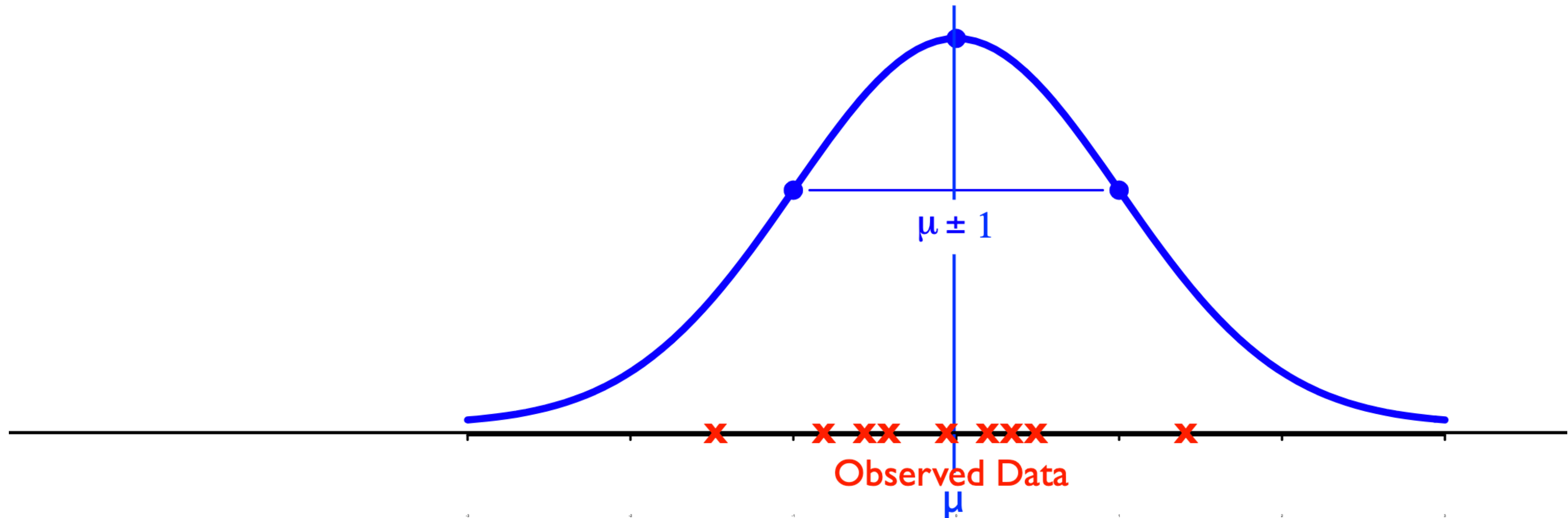
Which is more likely: (b) or this?

μ unknown, $\sigma^2 = 1$



Which is more likely: (c) or *this*?

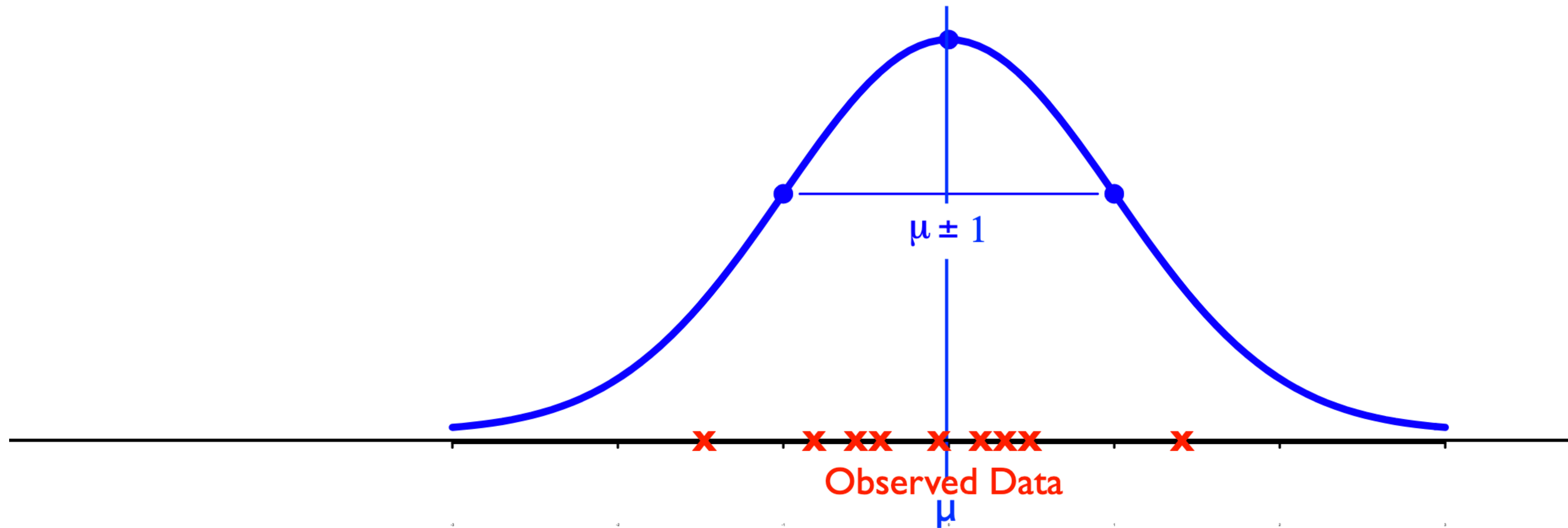
μ unknown, $\sigma^2 = 1$



Which is more likely: (c) or this?

μ unknown, $\sigma^2 = 1$

Looks good by eye, but how do I optimize my estimate of μ ?



Likelihood

(For Continuous Distributions)

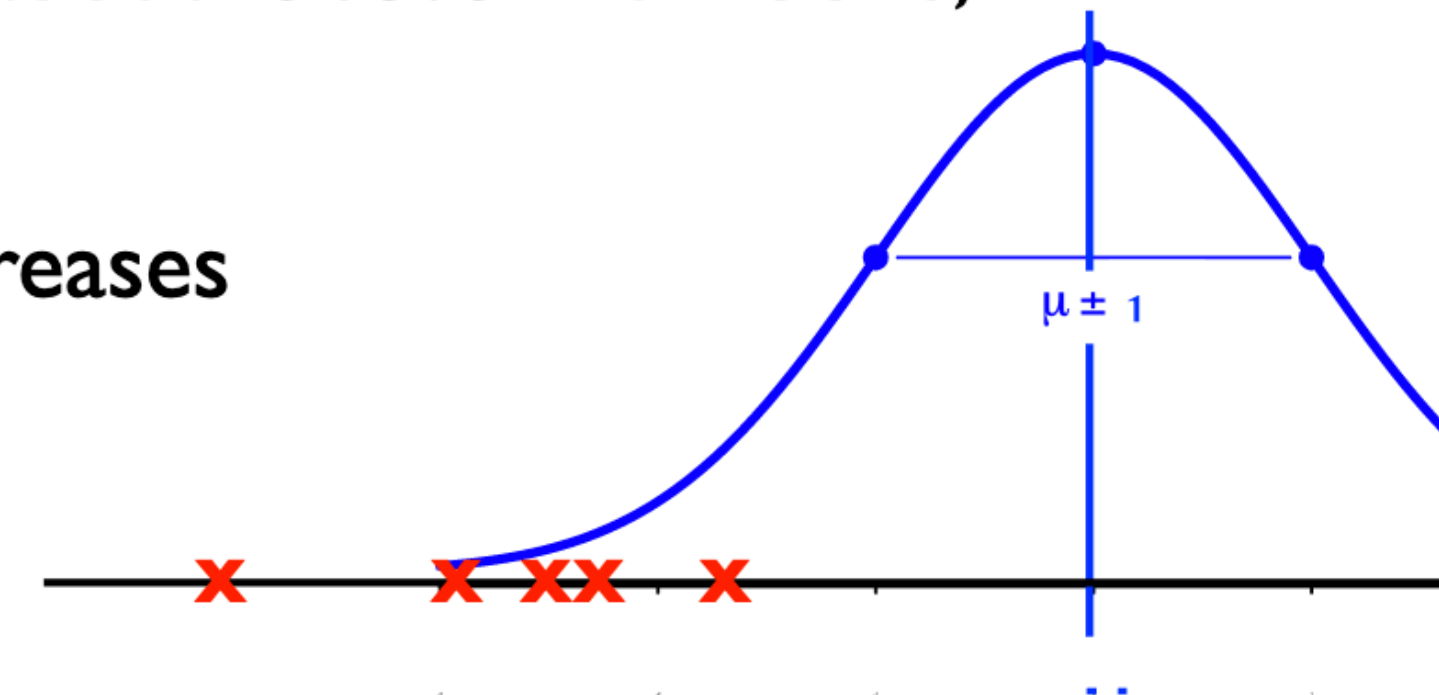
Probability of any specific observation x_i is zero, so “likelihood = probability” fails. Instead, as usual, we swap density for pmf: “likelihood” of $x_1, x_2, \dots, x_n$ is **defined** to be their *joint density*, and given independence of the x_i , that’s the product of their marginal densities. Why it’s sensible:

a) for maximizing likelihood, we really only care about *relative* likelihoods, and density captures that

b) it has the desired property that likelihood increases with better fit to the model

and

c) if density at x is $f(x)$, for any small $\delta > 0$, the probability of a sample within $\pm \delta/2$ of x is $\approx \delta f(x)$, so density really is capturing probability, and δ is *constant* wrt θ , so it just drops out of $d/d\theta \log L(\dots) = 0$.



Otherwise, MLE approach is just like discrete case: get likelihood, $\frac{\partial}{\partial \theta} \log L(\vec{x} | \theta) = 0$

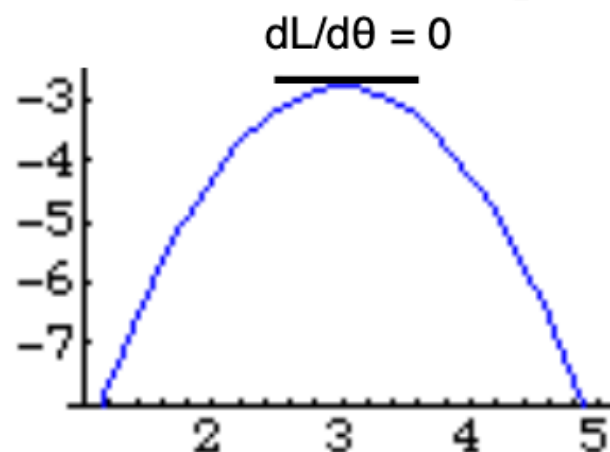
Ex. 2: $x_i \sim N(\mu, \sigma^2)$, $\sigma^2 = 1$, μ unknown

$$L(x_1, x_2, \dots, x_n | \theta) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}} e^{-(x_i - \theta)^2 / 2} \quad \leftarrow \text{product of densities}$$

$$\ln L(x_1, x_2, \dots, x_n | \theta) = \sum_{i=1}^n -\frac{1}{2} \ln(2\pi) - \frac{(x_i - \theta)^2}{2}$$

$$\frac{d}{d\theta} \ln L(x_1, x_2, \dots, x_n | \theta) = \sum_{i=1}^n (x_i - \theta)$$

And verify it's max,
not min & not better
on boundary

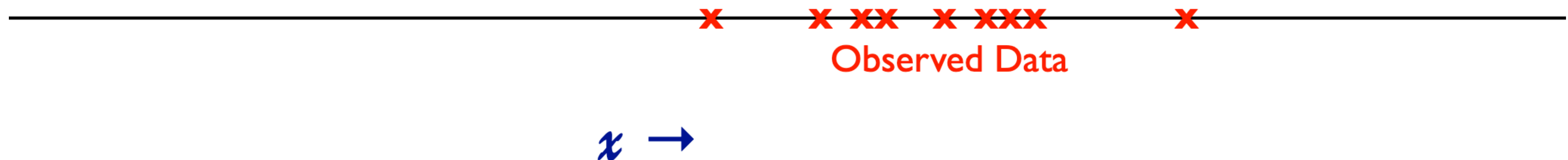


$$= \left(\sum_{i=1}^n x_i \right) - n\theta = 0$$

$$\hat{\theta} = \left(\sum_{i=1}^n x_i \right) / n = \bar{x}$$

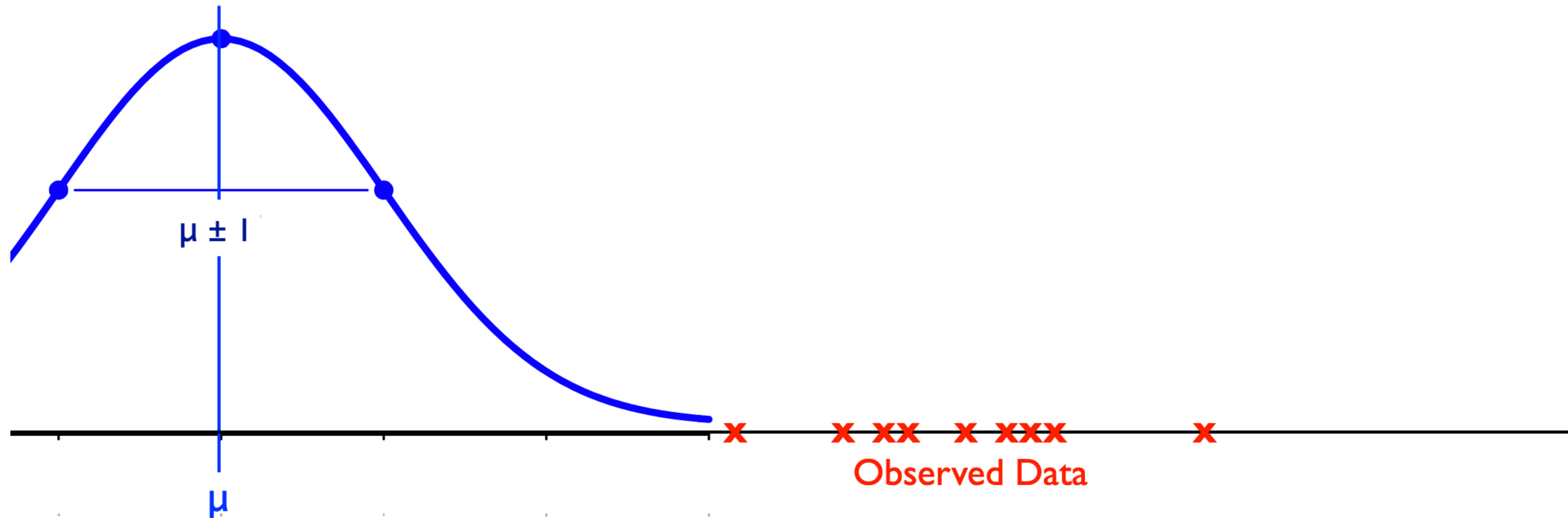
**Sample mean is MLE of
population mean**

Ex3: I got data; a little birdie tells me
it's normal (but does *not* tell me μ, σ^2)



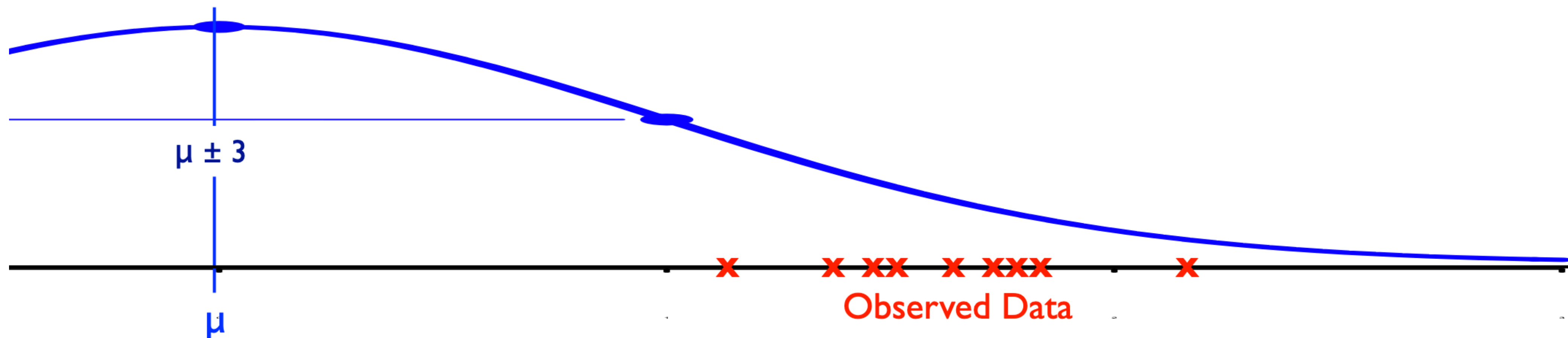
Which is more likely: (a) this?

μ, σ^2 both unknown



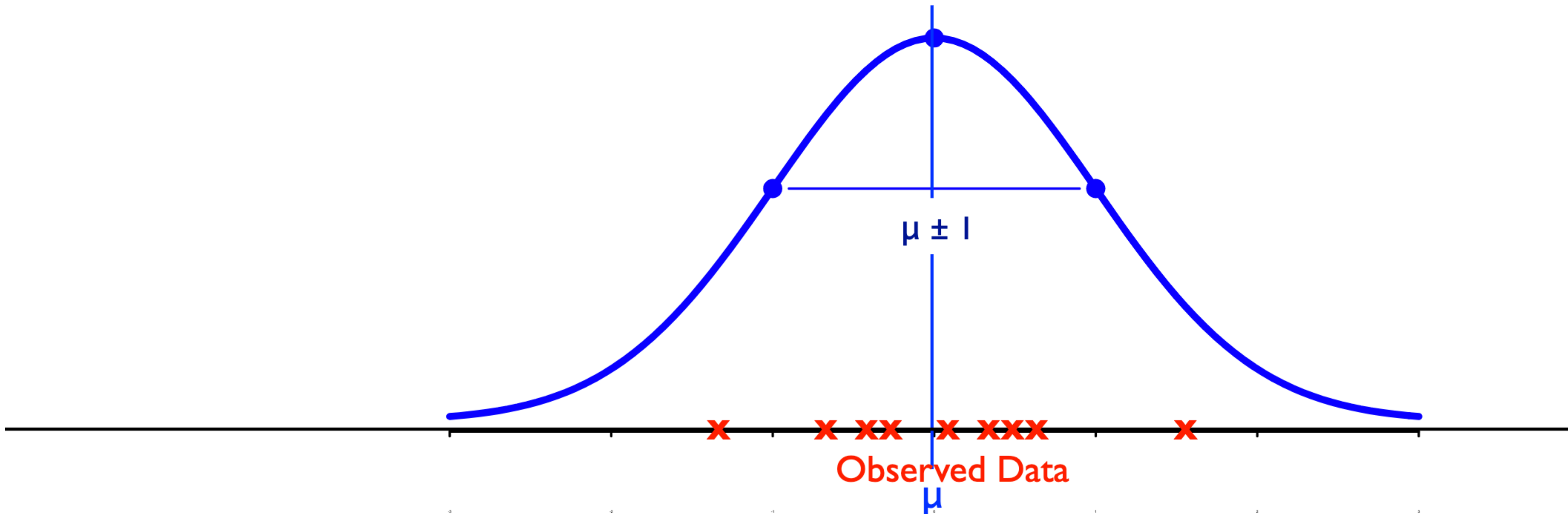
Which is more likely: (b) or this?

μ, σ^2 both unknown



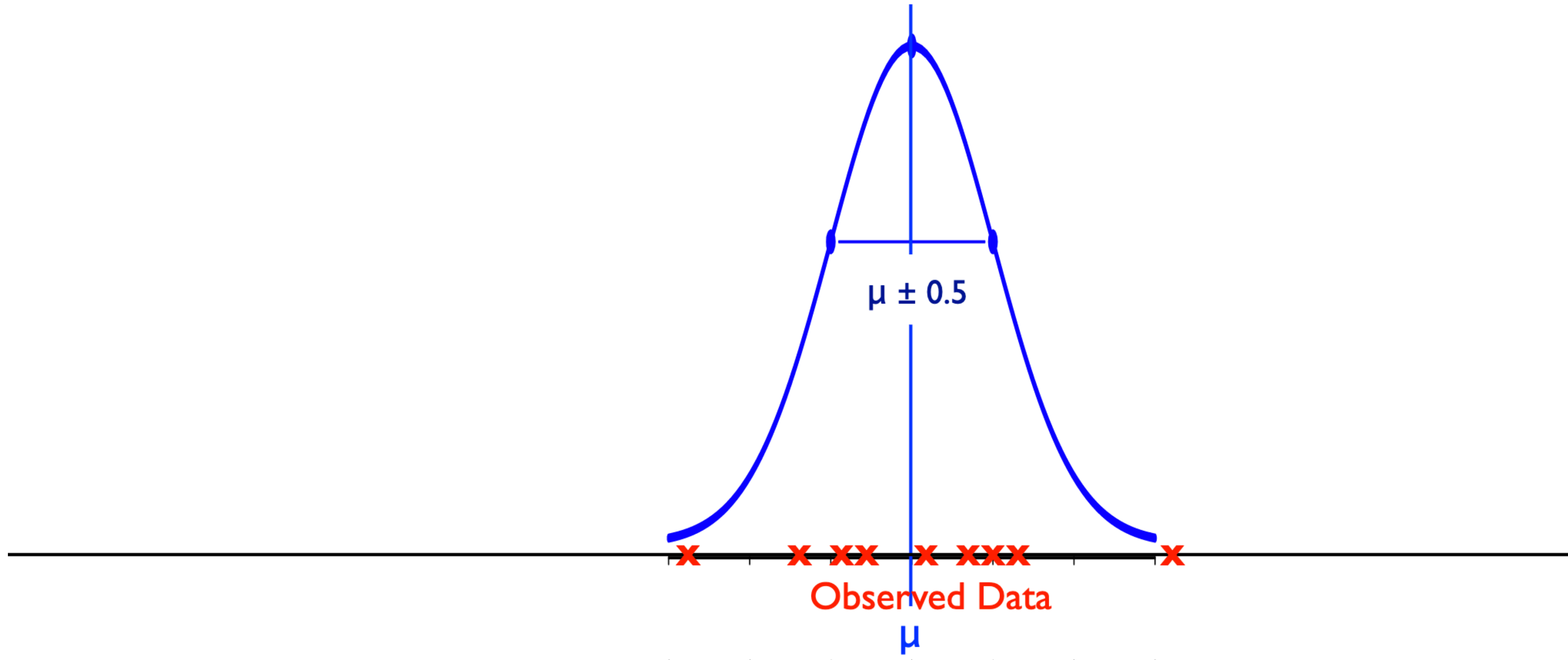
Which is more likely: (c) or this?

μ, σ^2 both unknown



Which is more likely: (d) or *this*?

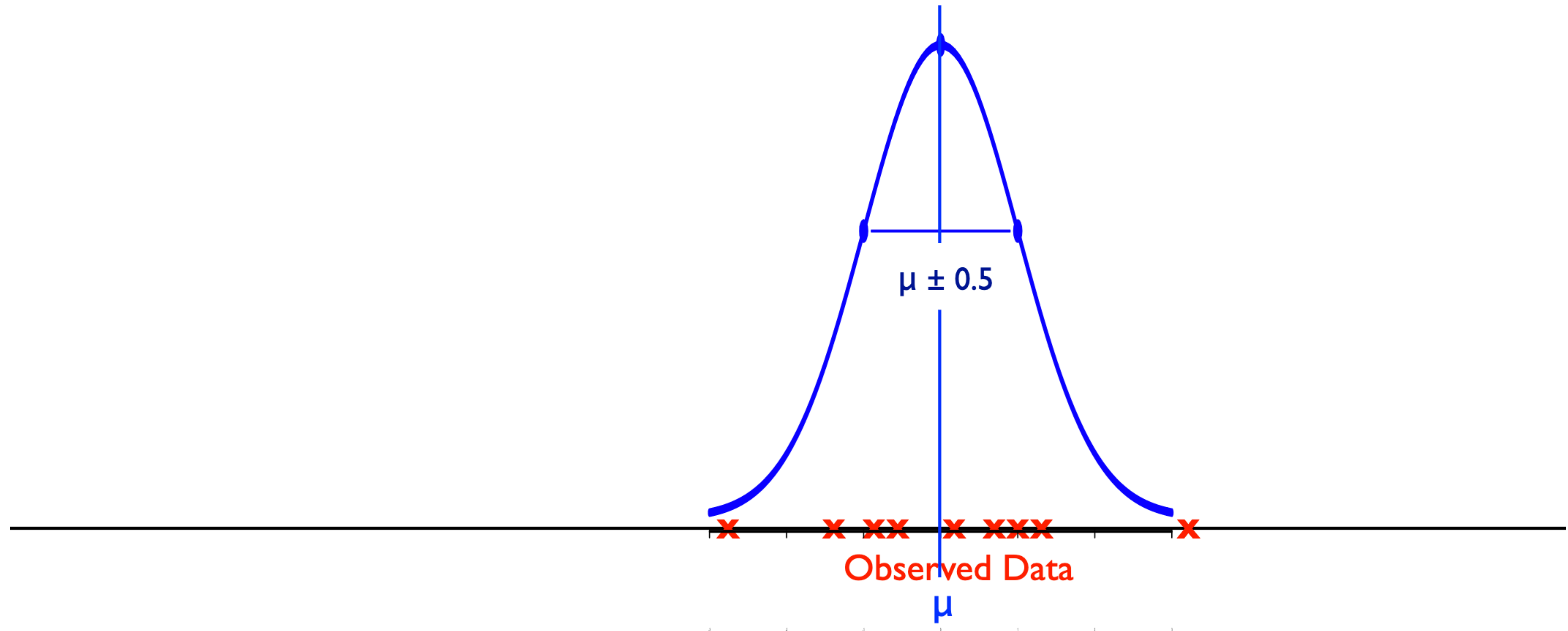
μ, σ^2 both unknown



Which is more likely: (d) or *this*?

μ, σ^2 both unknown

Looks good by eye, but how do I optimize my estimates of μ & $\underline{\underline{\sigma^2}}$?

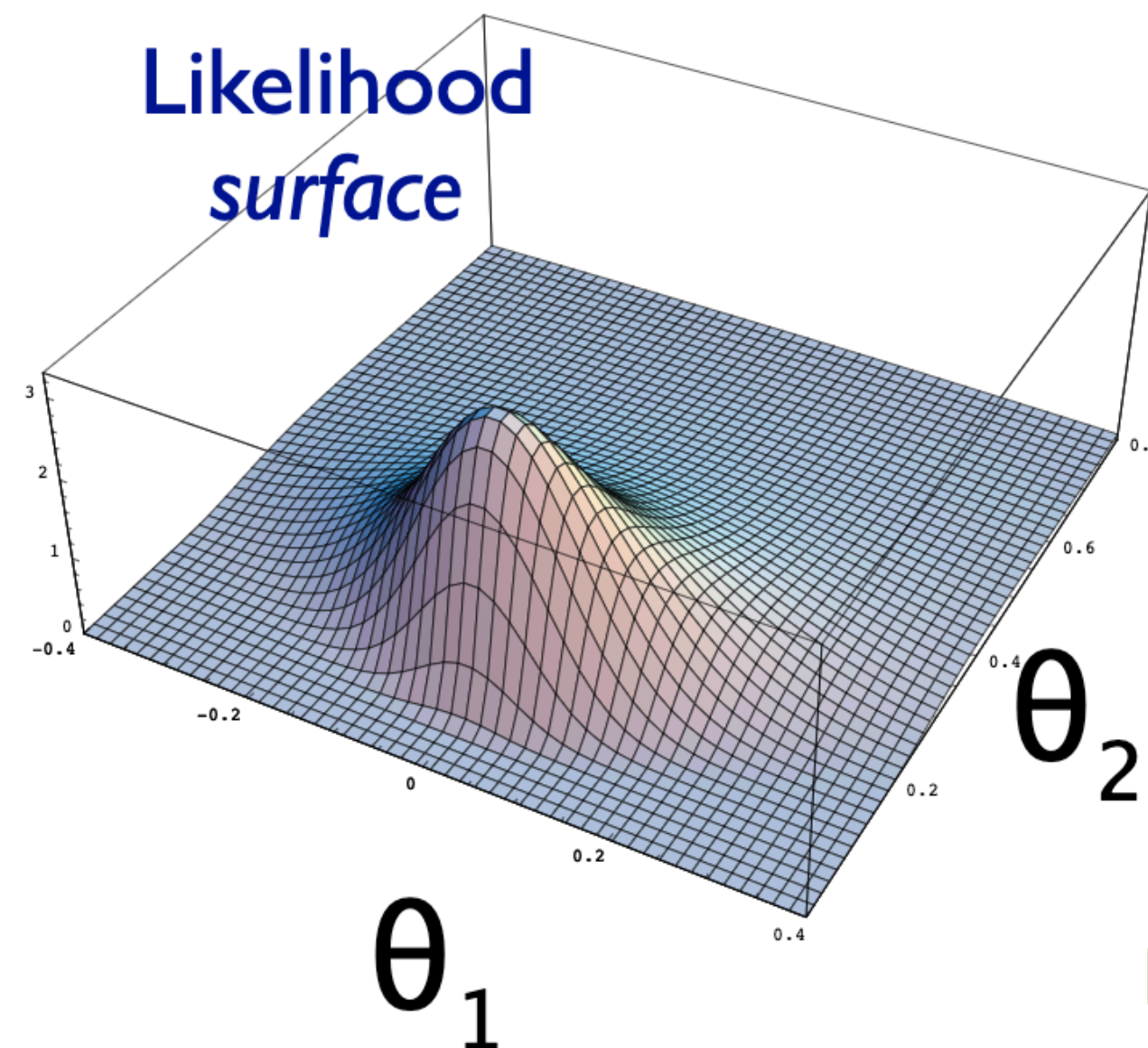


Ex 3: $x_i \sim N(\mu, \sigma^2)$, μ, σ^2 both unknown

$$\ln L(x_1, x_2, \dots, x_n | \theta_1, \theta_2) = \sum_{i=1}^n -\frac{1}{2} \ln(2\pi\theta_2) - \frac{(x_i - \theta_1)^2}{2\theta_2}$$

$$\frac{\partial}{\partial \theta_1} \ln L(x_1, x_2, \dots, x_n | \theta_1, \theta_2) = \sum_{i=1}^n \frac{(x_i - \theta_1)}{\theta_2} = 0$$

$$\hat{\theta}_1 = \left(\sum_{i=1}^n x_i \right) / n = \bar{x}$$



Sample mean is MLE of population mean, again

In general, a problem like this results in 2 equations in 2 unknowns. Easy in this case, since θ_2 drops out of the $\partial/\partial\theta_1 = 0$ equation

Ex. 3, (cont.)

$$\ln L(x_1, x_2, \dots, x_n | \theta_1, \theta_2) = \sum_{i=1}^n -\frac{1}{2} \ln(2\pi\theta_2) - \frac{(x_i - \theta_1)^2}{2\theta_2}$$

$$\frac{\partial}{\partial \theta_2} \ln L(x_1, x_2, \dots, x_n | \theta_1, \theta_2) = \sum_{i=1}^n -\frac{1}{2} \frac{2\pi}{2\pi\theta_2} + \frac{(x_i - \theta_1)^2}{2\theta_2^2} = 0$$

$$\hat{\theta}_2 = \left(\sum_{i=1}^n (x_i - \hat{\theta}_1)^2 \right) / n = \bar{s}^2$$

*Sample variance is MLE of
population variance*

Ex. 3, (cont.)

Bias? if Y is sample mean

$$Y = (\sum_{1 \leq i \leq n} X_i)/n$$

then

$$E[Y] = (\sum_{1 \leq i \leq n} E[X_i])/n = n \mu/n = \mu$$

so the MLE is an *unbiased* estimator of population mean

Similarly, $(\sum_{1 \leq i \leq n} (X_i - \mu)^2)/n$ is an unbiased estimator of σ^2 .

Unfortunately, if μ is unknown, estimated *from the same data*, as above, $\hat{\theta}_2 = \sum_{1 \leq i \leq n} \frac{(x_i - \hat{\theta}_1)^2}{n}$ is a consistent, but *biased* estimate of population variance. (An example of overfitting.) Unbiased estimate is:

$$\hat{\theta}'_2 = \sum_{1 \leq i \leq n} \frac{(x_i - \hat{\theta}_1)^2}{n-1}$$

i.e., $\lim_{n \rightarrow \infty}$
= correct

Summary MLE

MLE is *one* way to estimate *parameters* from *data*

You choose the *form* of the model (normal, binomial, ...)

Math chooses the *value(s)* of parameter(s)

Defining the “Likelihood Function” (based on the pmf or pdf of the model) is often the critical step; the math/algorithms to optimize it are generic

Often simply $(d/d\theta)(\log \text{Likelihood}(\text{data}|\theta)) = 0$

Has the intuitively appealing property that the parameters maximize the *likelihood* of the observed data; basically just assumes your sample is “representative”

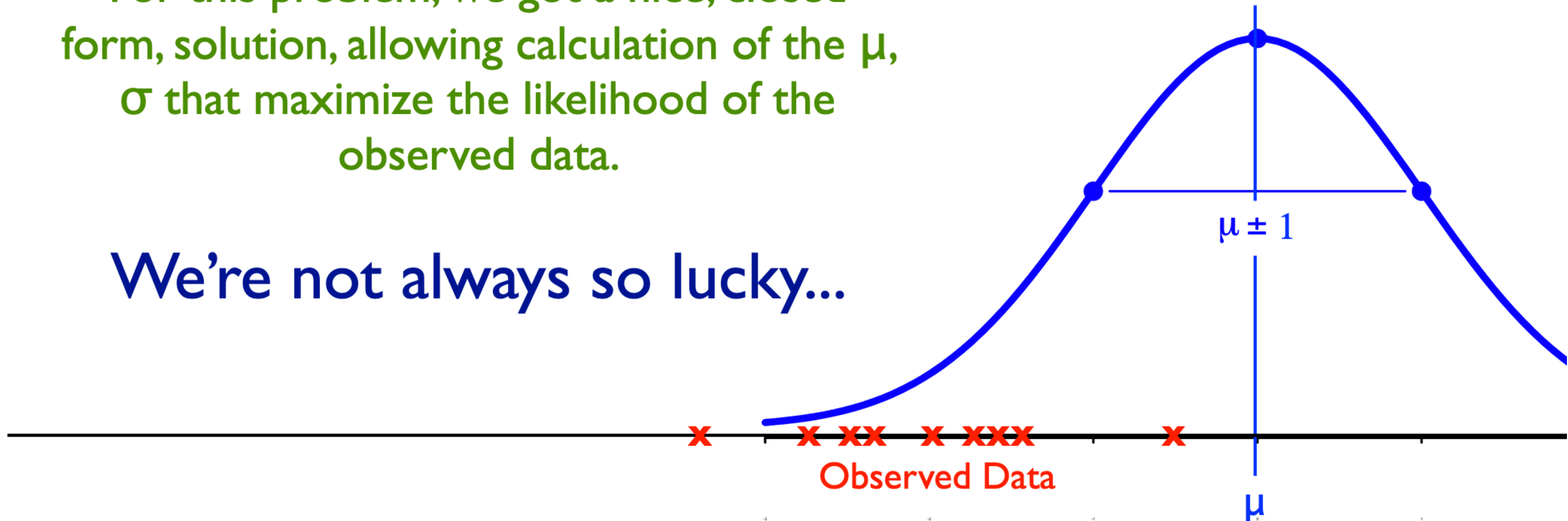
Of course, unusual samples will give bad estimates (estimate normal human heights from a sample of NBA stars?) but that is an unlikely event

Often, but not always, MLE has other desirable properties like being *unbiased*, or at least *consistent*

How to estimate μ given data

For this problem, we got a nice, closed form, solution, allowing calculation of the μ , σ that maximize the likelihood of the observed data.

We're not always so lucky...

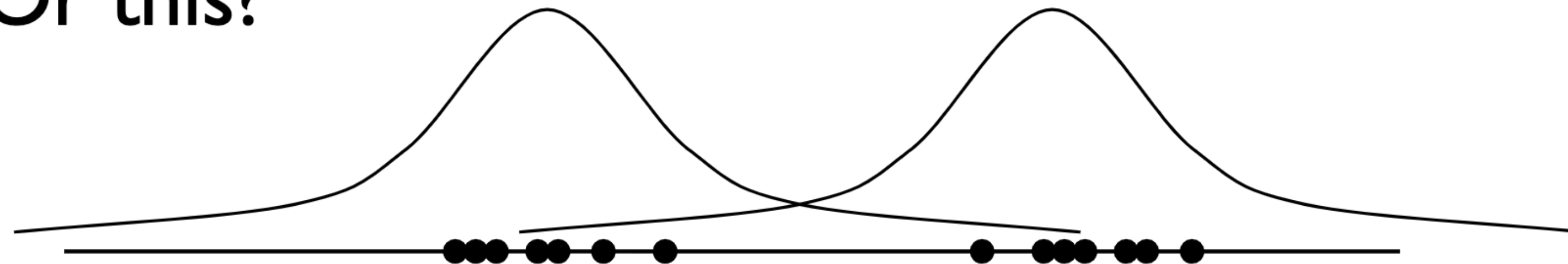
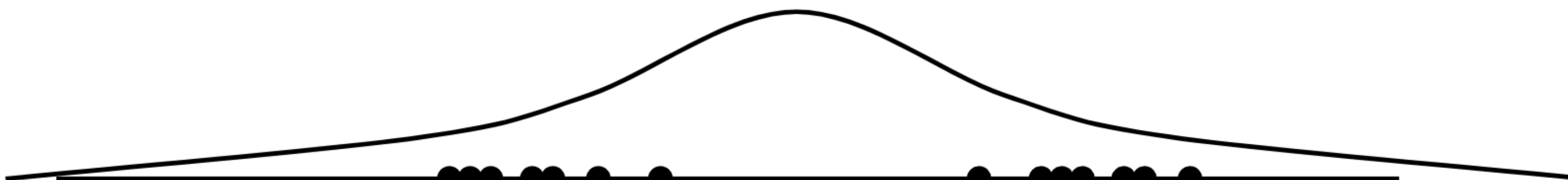


More Complex Example

This?

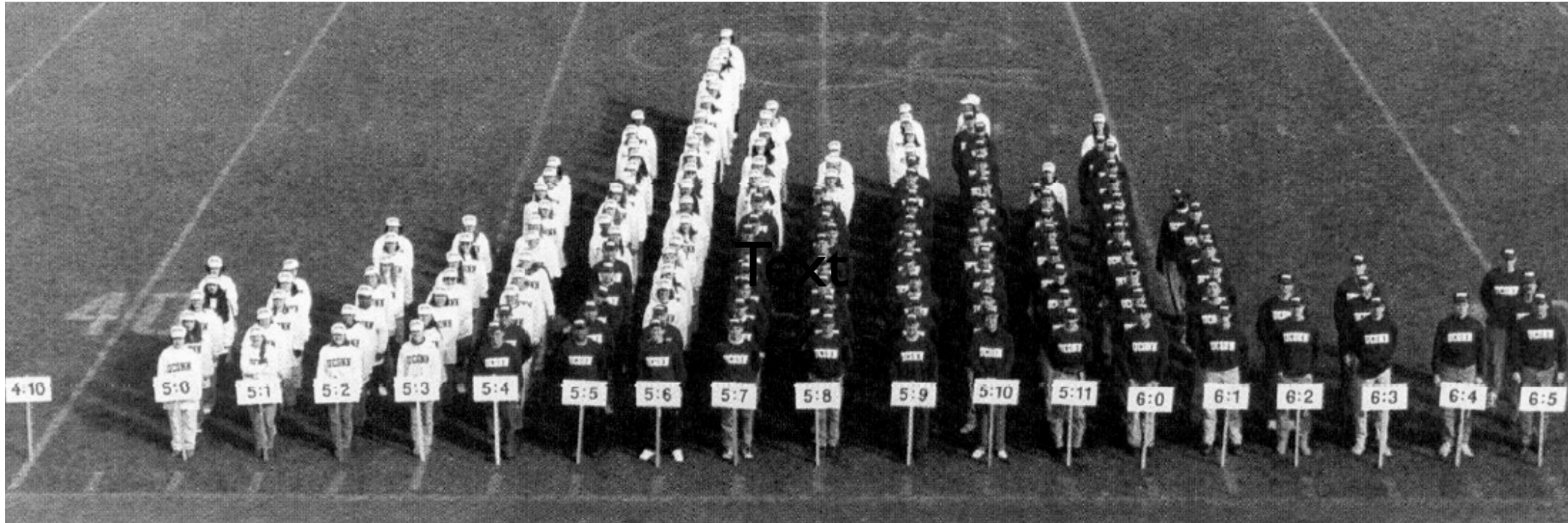


Or this?



(A modeling decision, not a math problem...,
but if the later, what math?)

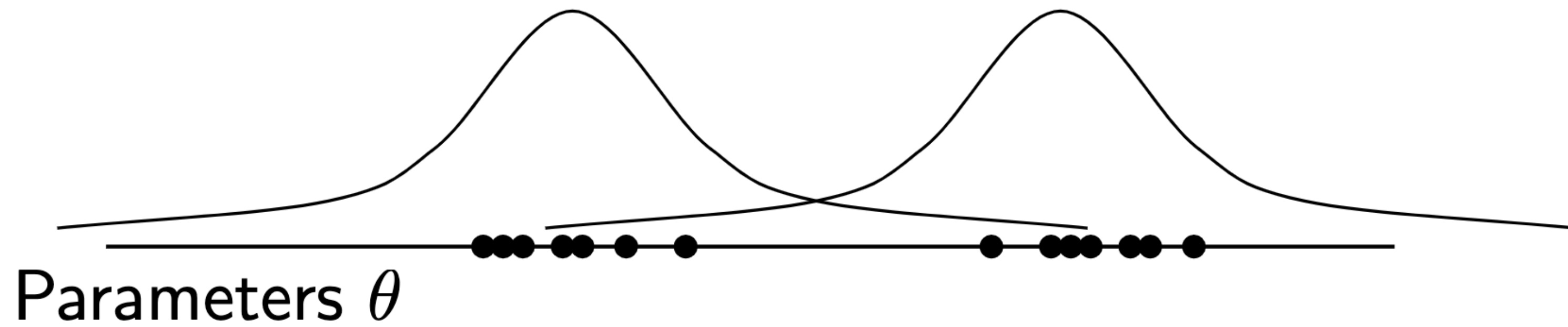
A Living Histogram



male and female genetics students, University of Connecticut in 1996

<http://mindprod.com/jgloss/histogram.html>

Gaussian Mixture Models / Model-based Clustering



means	μ_1	μ_2
variances	σ_1^2	σ_2^2
mixing parameters	τ_1	$\tau_2 = 1 - \tau_1$

P.D.F. $\xrightarrow{\text{separately}}$ $f(x|\mu_1, \sigma_1^2)$ $f(x|\mu_2, \sigma_2^2)$

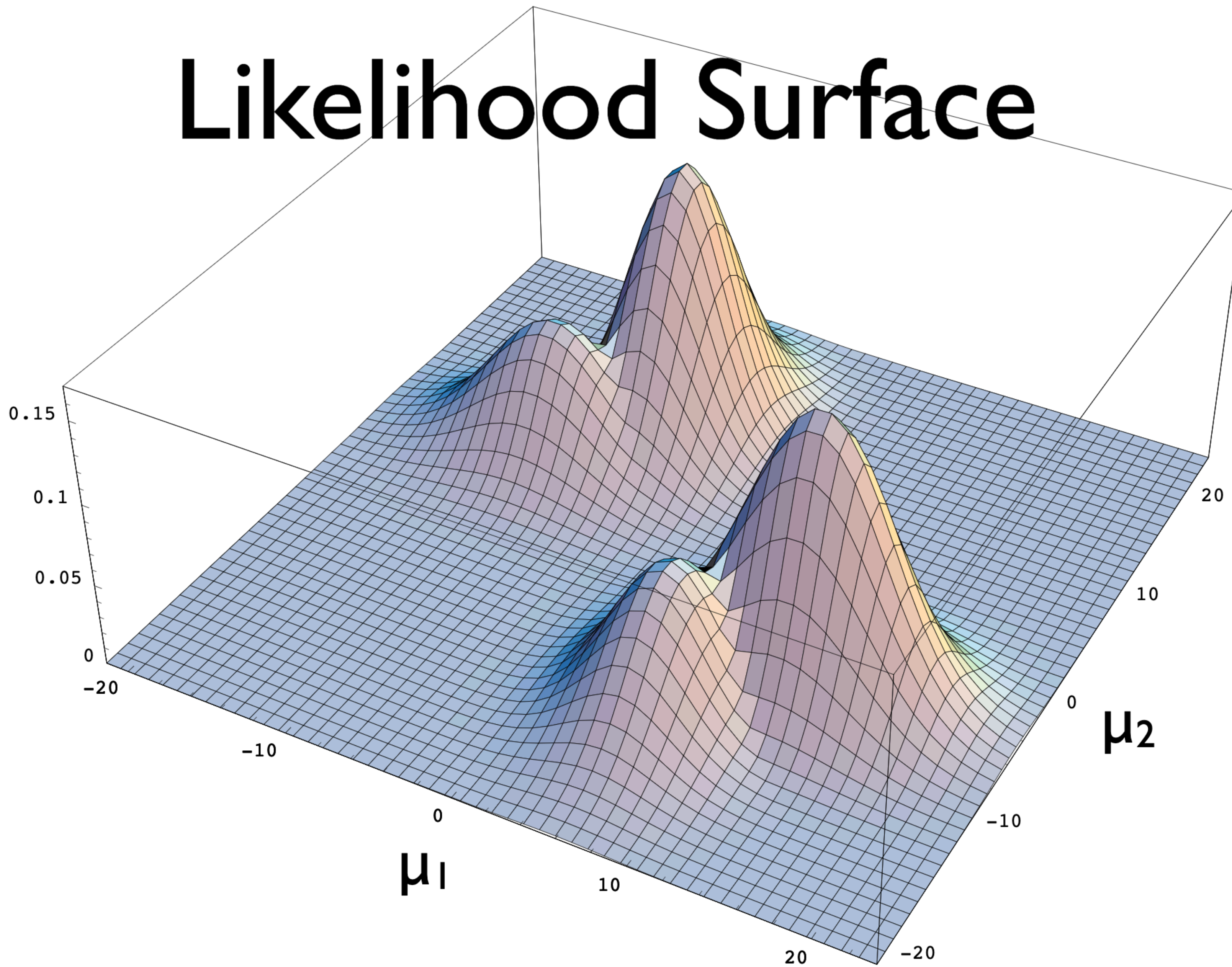
Likelihood $\xrightarrow{\text{together}}$ $\tau_1 f(x|\mu_1, \sigma_1^2) + \tau_2 f(x|\mu_2, \sigma_2^2)$

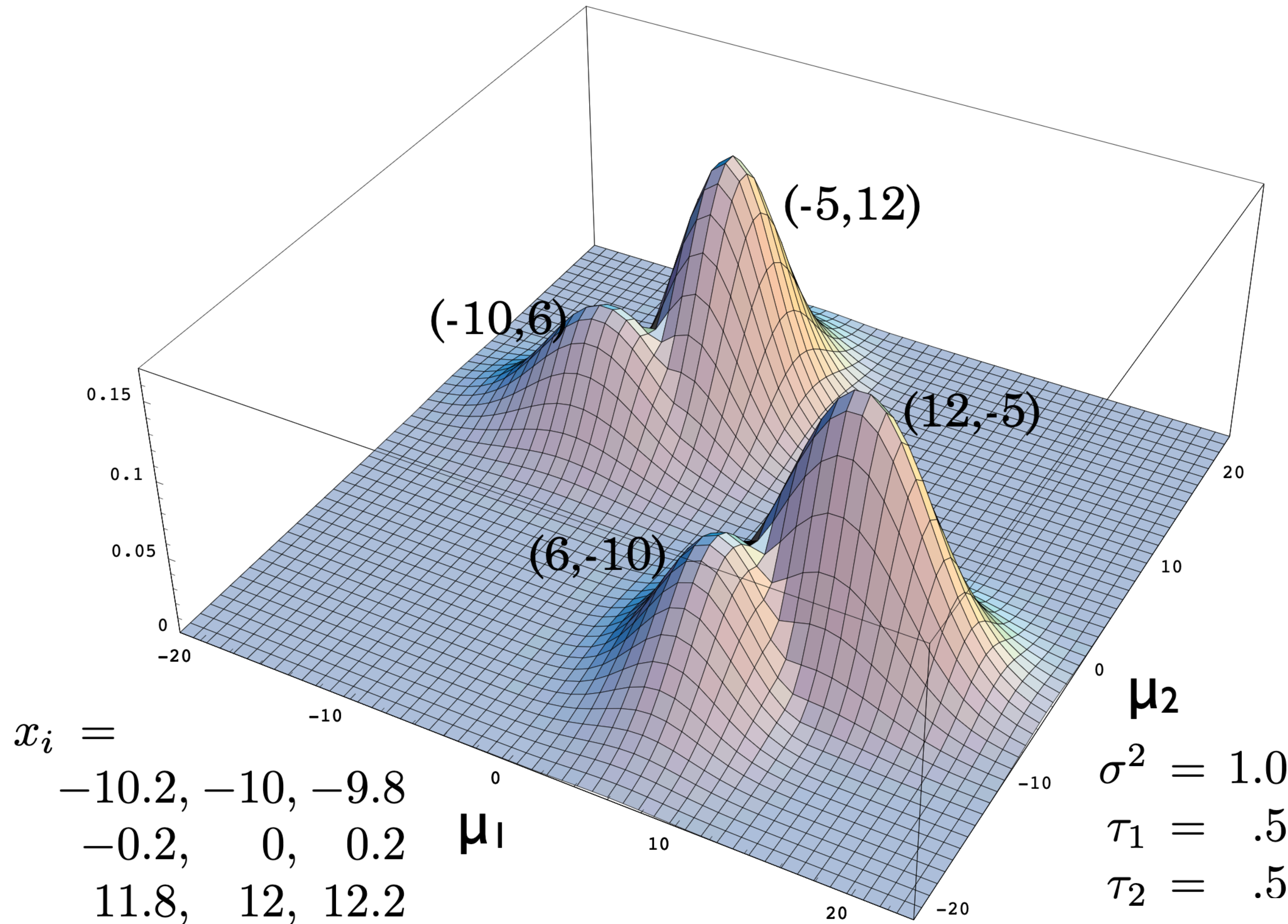
$$L(x_1, x_2, \dots, x_n | \mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \tau_1, \tau_2)$$

$$= \prod_{i=1}^n \sum_{j=1}^2 \tau_j f(x_i | \mu_j, \sigma_j^2)$$

No
closed-
form
max

Likelihood Surface





A What-If Puzzle

Likelihood

$$L(x_1, x_2, \dots, x_n | \overbrace{\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \tau_1, \tau_2}^{\theta})$$
$$= \prod_{i=1}^n \sum_{j=1}^2 \tau_j f(x_i | \mu_j, \sigma_j^2)$$

Messy: no closed form solution known for finding θ maximizing L

But *what if* we knew the *hidden data*?

$$z_{ij} = \begin{cases} 1 & \text{if } x_i \text{ drawn from } f_j \\ 0 & \text{otherwise} \end{cases}$$

A Hat Trick

Two slips of paper in a hat:

Pink: $\mu = 3$, and

Blue: $\mu = 7$.

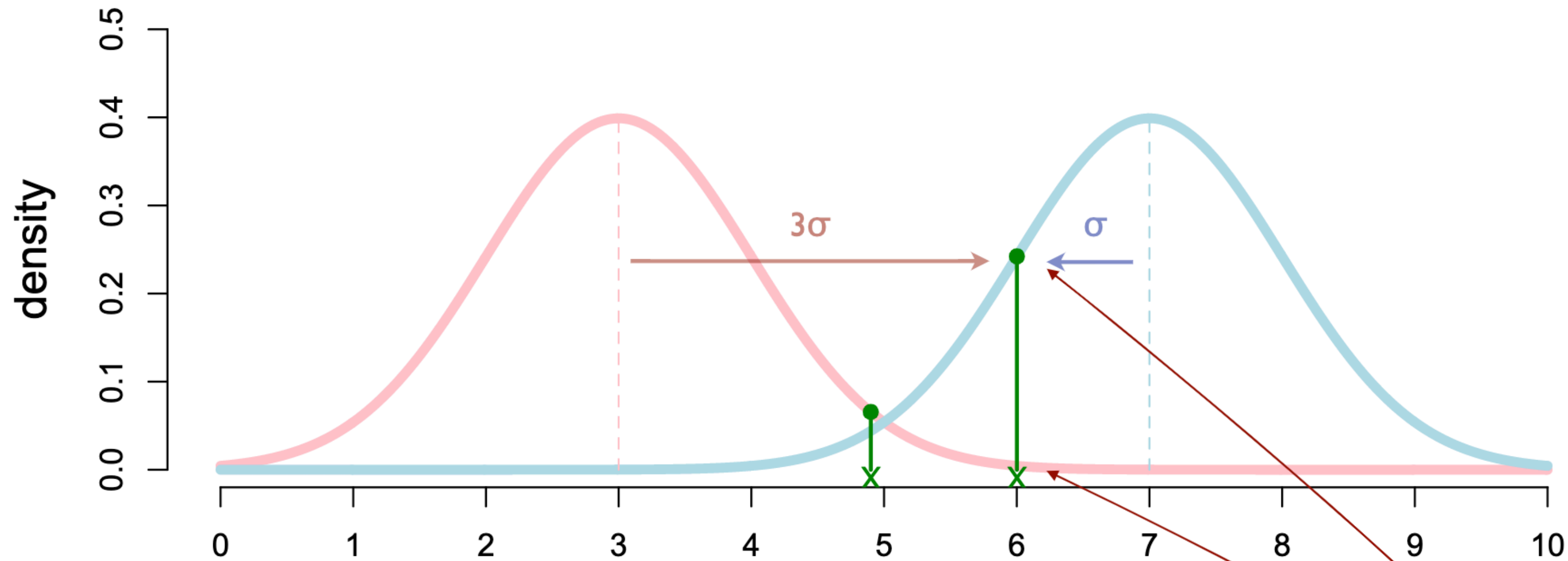
You draw one, then (without revealing color or μ)
reveal a single sample $X \sim \text{Normal}(\text{mean } \mu, \sigma^2 = 1)$.

You happen to draw $X = 6.001$.

Dr. Mean says “your slip = 7.” What is $P(\text{correct})$?

What if X had been 4.9?

A Hat Trick



Let “ $X \approx 6$ ” be a shorthand for $6.001 - \delta/2 < X < 6.001 + \delta/2$

$$P(\mu = 7|X = 6) = \lim_{\delta \rightarrow 0} P(\mu = 7|X \approx 6)$$

$$P(\mu = 7|X \approx 6) = \frac{P(X \approx 6|\mu = 7)P(\mu = 7)}{P(X \approx 6)} \quad \text{Bayes rule}$$

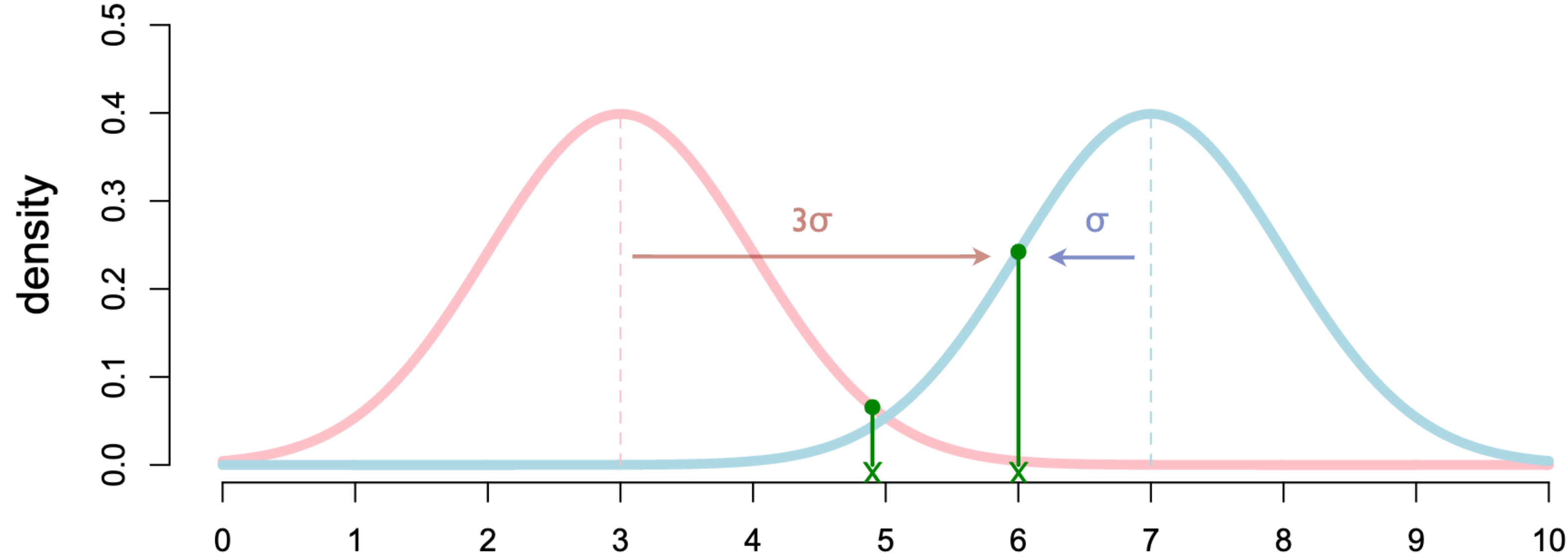
$$= \frac{0.5P(X \approx 6|\mu = 7)}{0.5P(X \approx 6|\mu = 3) + 0.5P(X \approx 6|\mu = 7)}$$

$$\approx \frac{f(X = 6|\mu = 7)\delta}{f(X = 6|\mu = 3)\delta + f(X = 6|\mu = 7)\delta}, \text{ so}$$

$$P(\mu = 7|X = 6) = \frac{f(X = 6|\mu = 7)}{f(X = 6|\mu = 3) + f(X = 6|\mu = 7)} \approx 0.982$$

f = normal
density

A Hat Trick



Alternate View:

f = normal
density

Posterior odds = Bayes Factor · Prior odds

$$\frac{P(\mu = 7|X = 6)}{P(\mu = 3|X = 6)} = \frac{f(X = 6|\mu = 7)}{f(X = 6|\mu = 3)} \cdot \frac{0.50}{0.50} = \frac{0.2422}{0.0044} \cdot \frac{1}{1} = \frac{54.8}{1}$$

I.e., 50:50 prior odds become 54:1 in favor of $\mu=7$, given $X=6.00$!

(and would become 3:2 in favor of $\mu=3$, given $X=4.9$)

Another Hat Trick

Two secret numbers, μ_{pink} and μ_{blue}

On pink slips, many samples of $\text{Normal}(\mu_{\text{pink}}, \sigma^2 = 1)$,

Ditto on blue slips, from $\text{Normal}(\mu_{\text{blue}}, \sigma^2 = 1)$.

Based on 16 of each, how would you “guess” the secrets (where “success” means your guess is within ± 0.5 of each secret)?

Roughly how likely is it that you will succeed?

Another Hat Trick (cont.)

Pink/blue = red herrings; separate & independent

Given $X_1, \dots, X_{16} \sim N(\mu, \sigma^2)$, $\sigma^2 = 1$

Calculate $Y = (X_1 + \dots + X_{16})/16 \sim N(?, ?)$

$E[Y] = \mu$

$\text{Var}(Y) = 16\sigma^2/16^2 = \sigma^2/16 = 1/16$

I.e., X_i 's are all $\sim N(\mu, 1)$; Y is $\sim N(\mu, 1/16)$

and since $0.5 = 2 \sqrt{1/16}$, we have:

“ Y within ± 0.5 of μ ” = “ Y within $\pm 2 \sigma$ of μ ” $\approx 95\%$ prob

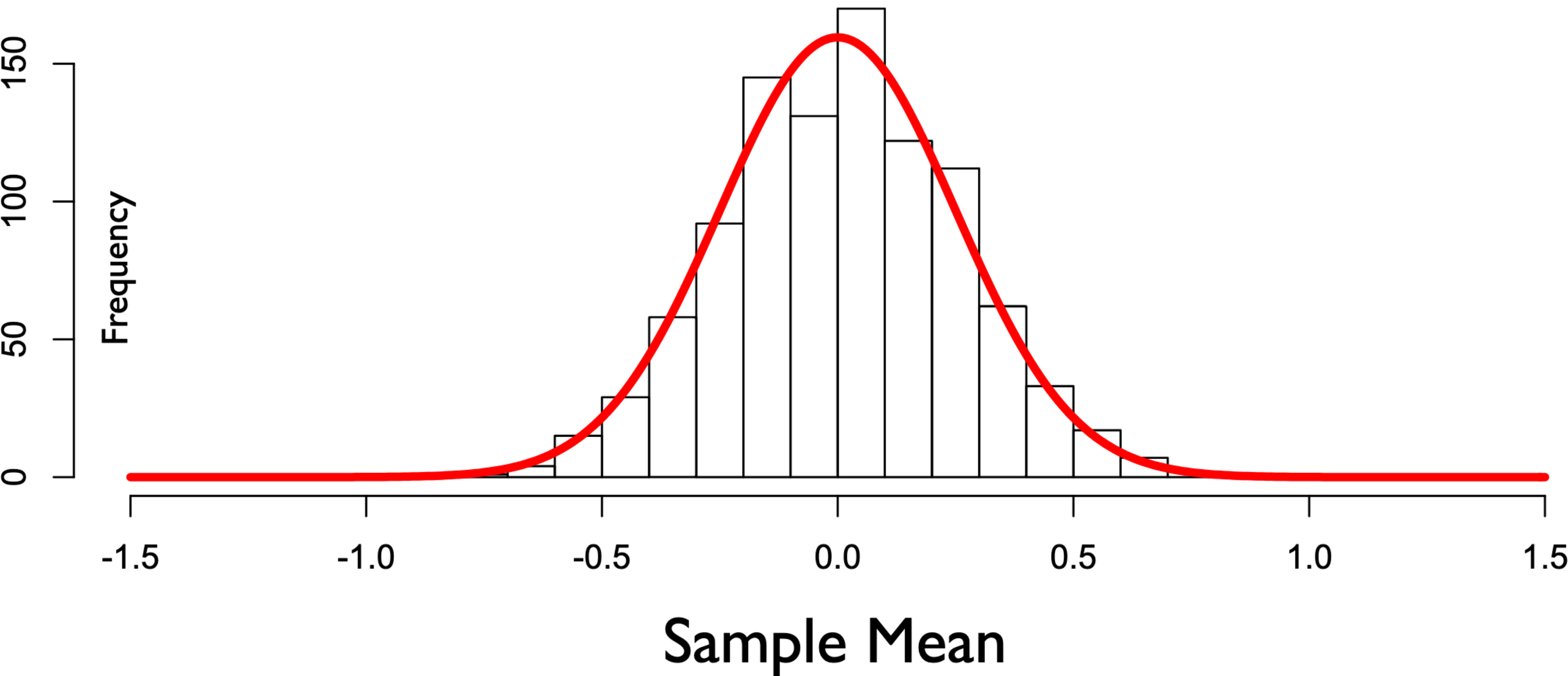
Note 1: Y is a *point estimate* for μ ;

$Y \pm 2 \sigma$ is a *95% confidence interval* for μ

(More on this topic later)

Histogram of 1000 samples of the average of 16 N(0,1) RVs

Red = $N(0, 1/16)$ density

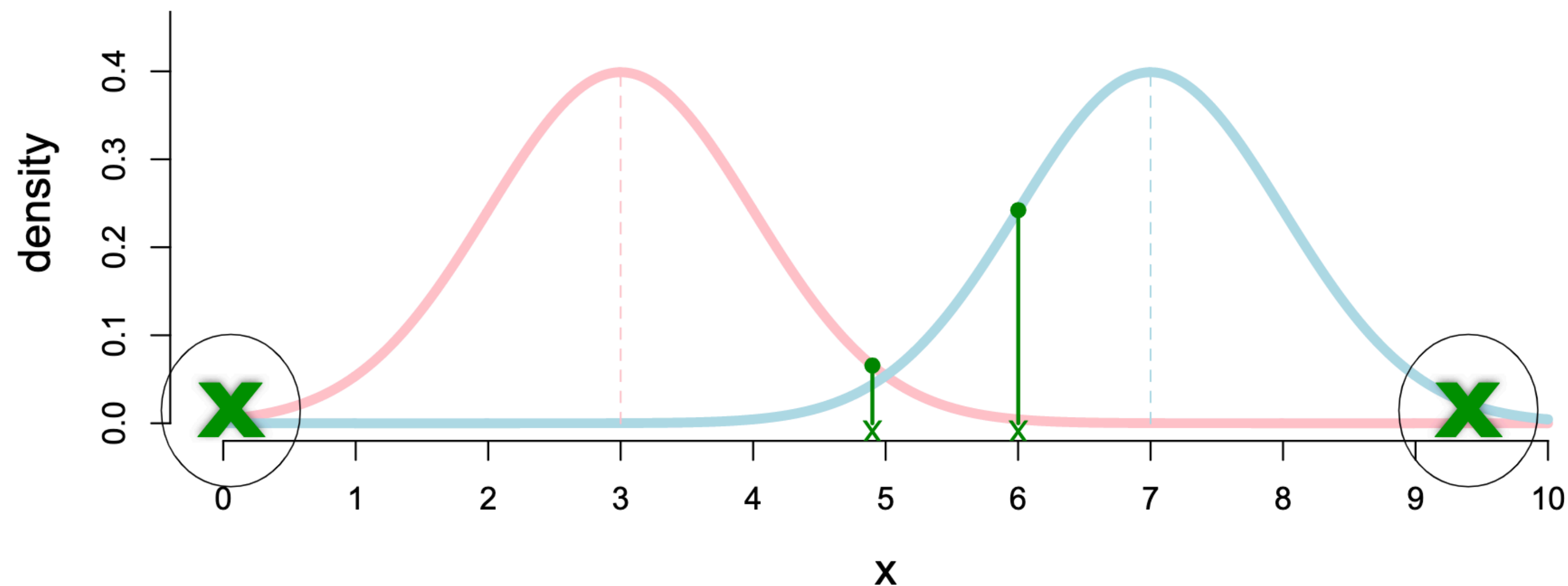


Hat Trick 2 (cont.)

Note 2:

What would you do if some of the slips you pulled had coffee spilled on them, obscuring color?

If they were half way between means of the others?
If they were on opposite sides of the means of the others



A What-If Puzzle

Likelihood

$$L(x_1, x_2, \dots, x_n \mid \overbrace{\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \tau_1, \tau_2}^{\theta})$$
$$= \prod_{i=1}^n \sum_{j=1}^2 \tau_j f(x_i \mid \mu_j, \sigma_j^2)$$

Messy: no closed form solution known for finding θ maximizing L

But *what if* we knew the *hidden data*?

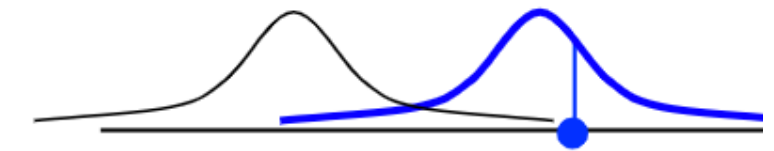
$$z_{ij} = \begin{cases} 1 & \text{if } x_i \text{ drawn from } f_j \\ 0 & \text{otherwise} \end{cases}$$

EM as Egg vs Chicken

Hat
Trick 1

IF parameters θ known, could estimate z_{ij}

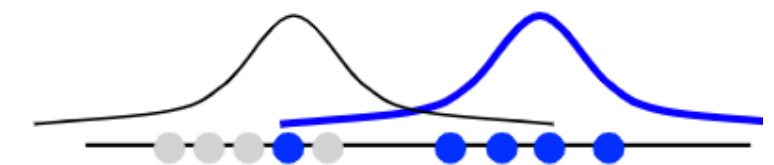
E.g., $|x_i - \mu_1|/\sigma_1 \gg |x_i - \mu_2|/\sigma_2 \Rightarrow P[z_{i1}=1] \ll P[z_{i2}=1]$



Hat
Trick 2

IF z_{ij} known, could estimate parameters θ

E.g., only points in cluster 2 influence μ_2, σ_2



But we know neither; (optimistically) iterate:

Hat
Trick 1

E-step: calculate expected z_{ij} , given parameters

Hat
Trick 2

M-step: calculate “MLE” of parameters, given $E(z_{ij})$

Overall, a clever “hill-climbing” strategy

Simple Version: “Classification EM”

If $E[z_{ij}] < .5$, pretend $z_{ij} = 0$; $E[z_{ij}] > .5$, pretend it's 1

I.e., *classify* points as component 1 or 2

Now recalc θ , assuming that partition (standard MLE)

Then recalc $E[z_{ij}]$, assuming that θ

Then re-recalc θ , assuming new $E[z_{ij}]$, etc., etc.

“Full EM” is slightly more involved, (to account for uncertainty in classification) but this is the crux.

“K-means
clustering,”
essentially

Full EM

x_i 's are known; θ unknown. Goal is to find MLE θ of:

$$L(x_1, \dots, x_n \mid \theta) \quad \text{(hidden data likelihood)}$$

Would be easy *if* z_{ij} 's were known, i.e., consider:

$$L(x_1, \dots, x_n, z_{11}, z_{12}, \dots, z_{n2} \mid \theta) \quad \text{(complete data likelihood)}$$

But z_{ij} 's aren't known.

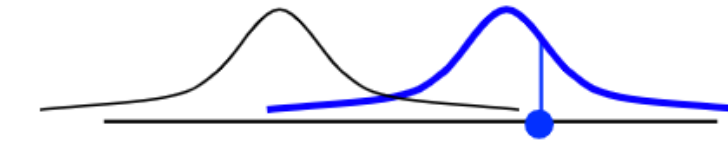
Instead, maximize *expected* likelihood of visible data

$$E(L(x_1, \dots, x_n, z_{11}, z_{12}, \dots, z_{n2} \mid \theta)),$$

where expectation is over distribution of hidden data (z_{ij} 's)

The E-step:

Find $E(z_{ij})$, i.e., $P(z_{ij}=1)$



Assume θ known & fixed

A (B): the event that x_i was drawn from f_1 (f_2)

D: the observed datum x_i

Expected value of z_{i1} is $P(A|D)$

$$E = 0 \cdot P(0) + 1 \cdot P(1)$$

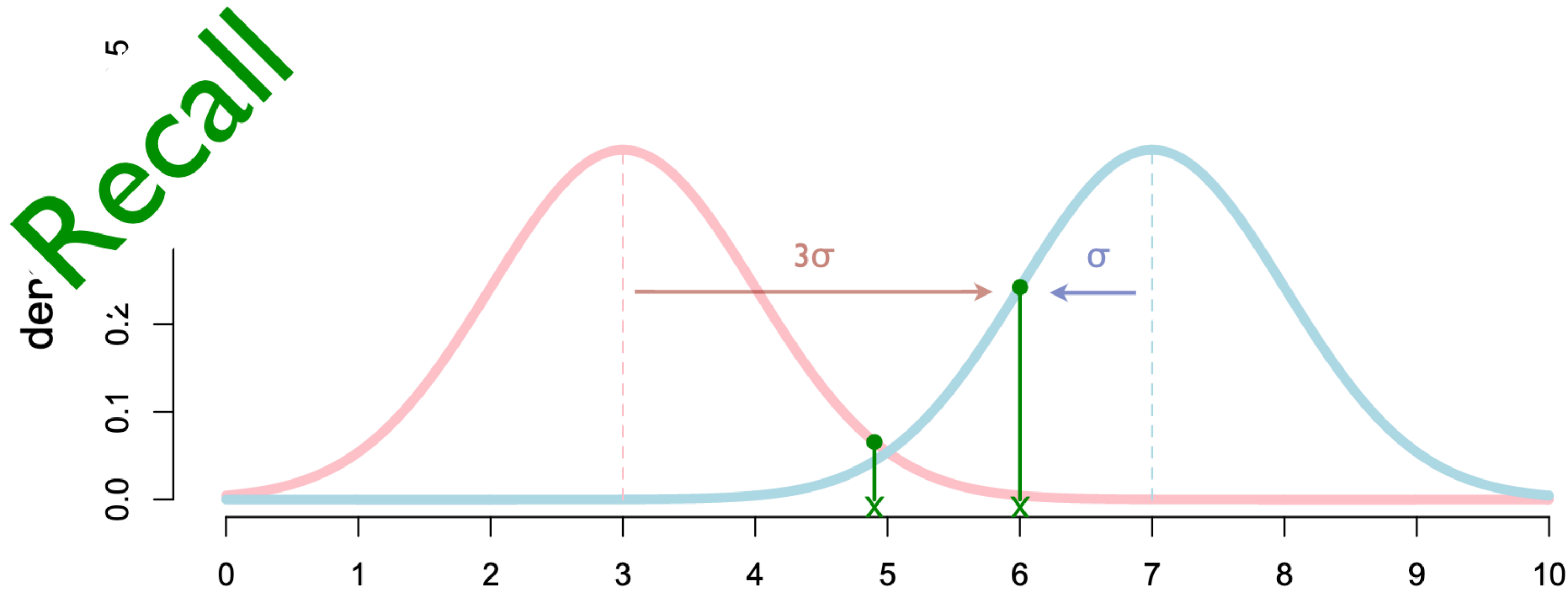
$$E[z_{i1}] = P(A|D) = \frac{P(D|A)P(A)}{P(D)}$$

$$\begin{aligned} P(D) &= P(D|A)P(A) + P(D|B)P(B) \\ &= f_1(x_i|\theta_1)\tau_1 + f_2(x_i|\theta_2)\tau_2 \end{aligned}$$

Repeat
for
each
 $z_{i,j}$

Note: denominator = sum of numerators – i.e. that which normalizes sum to 1 (typical Bayes)

A Hat Trick



Let “ $X \approx 6$ ” be a shorthand for $6.001 - \delta/2 < X < 6.001 + \delta/2$

$$P(\mu = 7|X = 6) = \lim_{\delta \rightarrow 0} P(\mu = 7|X \approx 6)$$

$$P(\mu = 7|X \approx 6) = \frac{P(X \approx 6|\mu = 7)P(\mu = 7)}{P(X \approx 6)}$$

$$= \frac{0.5P(X \approx 6|\mu = 7)}{0.5P(X \approx 6|\mu = 3) + 0.5P(X \approx 6|\mu = 7)}$$

$$\approx \frac{f(X = 6|\mu = 7)\delta}{f(X = 6|\mu = 3)\delta + f(X = 6|\mu = 7)\delta}, \text{ so}$$

$$P(\mu = 7|X = 6) = \frac{f(X = 6|\mu = 7)}{f(X = 6|\mu = 3) + f(X = 6|\mu = 7)} \approx 0.982$$

$E[Z_{i,\text{pink}}] = ?$

$E[Z_{i,\text{blue}}] =$

f = normal density

Complete Data Likelihood

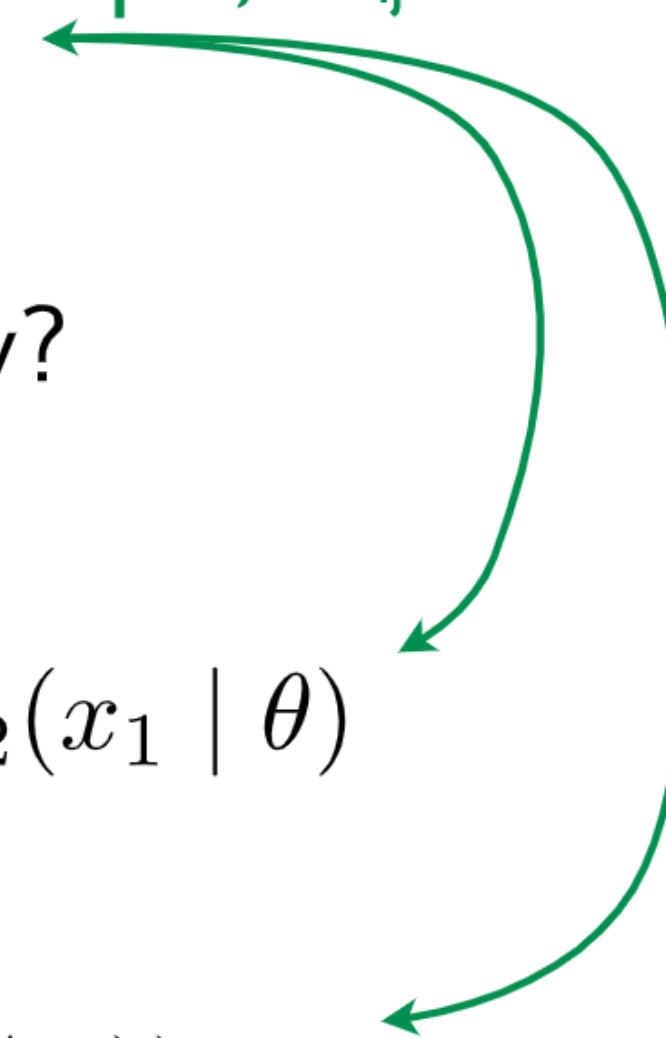
Recall:

$$z_{1j} = \begin{cases} 1 & \text{if } x_1 \text{ drawn from } f_j \\ 0 & \text{otherwise} \end{cases}$$

so, correspondingly,

$$L(x_1, z_{1j} \mid \theta) = \begin{cases} \tau_1 f_1(x_1 \mid \theta) & \text{if } z_{11} = 1 \\ \tau_2 f_2(x_1 \mid \theta) & \text{otherwise} \end{cases}$$

equal, if z_{ij} are 0/1



Formulas with “if’s” are messy; can we blend more smoothly?

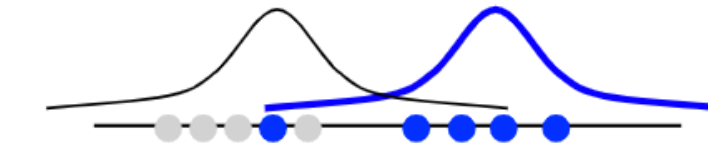
Yes, many possibilities. Idea 1:

$$L(x_1, z_{1j} \mid \theta) = z_{11} \cdot \tau_1 f_1(x_1 \mid \theta) + z_{12} \cdot \tau_2 f_2(x_1 \mid \theta)$$

Idea 2 (Better):

$$L(x_1, z_{1j} \mid \theta) = (\tau_1 f_1(x_1 \mid \theta))^{z_{11}} \cdot (\tau_2 f_2(x_1 \mid \theta))^{z_{12}}$$

M-step:



Find θ maximizing $E(\log(\text{Likelihood}))$

(For simplicity, assume $\sigma_1 = \sigma_2 = \sigma$; $\tau_1 = \tau_2 = \tau = 0.5$)

$$L(\vec{x}, \vec{z} \mid \theta) = \prod_{i=1}^n \left(\frac{\tau}{\sqrt{2\pi\sigma^2}} \exp \left(- \sum_{j=1}^2 z_{ij} \frac{(x_i - \mu_j)^2}{2\sigma^2} \right) \right)$$

$$\begin{aligned} E[\log L(\vec{x}, \vec{z} \mid \theta)] &= E \left[\sum_{i=1}^n \left(\log \tau - \frac{1}{2} \log(2\pi\sigma^2) - \sum_{j=1}^2 z_{ij} \frac{(x_i - \mu_j)^2}{2\sigma^2} \right) \right] \\ &= \sum_{i=1}^n \left(\log \tau - \frac{1}{2} \log(2\pi\sigma^2) - \sum_{j=1}^2 E[z_{ij}] \frac{(x_i - \mu_j)^2}{2\sigma^2} \right) \end{aligned}$$

wrt dist of z_{ij}

Find θ maximizing this as before, using $E[z_{ij}]$ found in E-step. Result:

$$\mu_j = \sum_{i=1}^n E[z_{ij}] x_i / \sum_{i=1}^n E[z_{ij}] \quad (\text{intuit: avg, weighted by subpop prob})$$

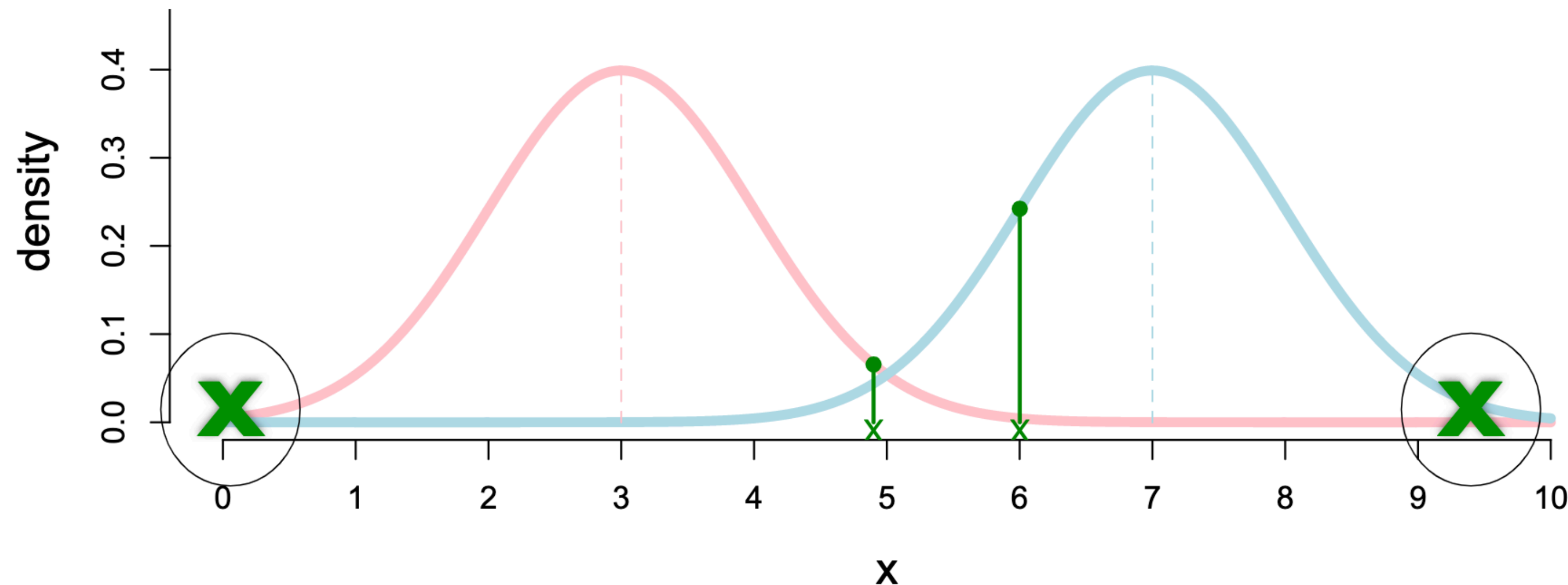
Recall

Hat Trick 2 (cont.)

Note 2: red/blue separation is just like the M-step of EM if values of the hidden variables (z_{ij}) were known.

What if they're not? E.g., what would you do if some of the slips you pulled had coffee spilled on them, obscuring color?

If they were half way between means of the others?
If they were on opposite sides of the means of the others



2 Component Mixture

$$\sigma_1 = \sigma_2 = 1; \tau = 0.5$$

			mu1	-20.00		-6.00		-5.00		-4.99
			mu2	6.00		0.00		3.75		3.75
x1	-6		z11		5.11E-12		1.00E+00		1.00E+00	
x2	-5		z21		2.61E-23		1.00E+00		1.00E+00	
x3	-4		z31		1.33E-34		9.98E-01		1.00E+00	
x4	0		z41		9.09E-80		1.52E-08		4.11E-03	
x5	4		z51		6.19E-125		5.75E-19		2.64E-18	
x6	5		z61		3.16E-136		1.43E-21		4.20E-22	
x7	6		z71		1.62E-147		3.53E-24		6.69E-26	

Essentially converged in 2 iterations

EM Summary

Fundamentally a maximum likelihood parameter estimation problem; broader than just Gaussian

Useful if 0/1 hidden data, and if analysis would be more tractable if hidden data z were known

Iterate:

E-step: estimate $E(z)$ for each z , given θ

M-step: estimate θ maximizing $E[\log \text{likelihood}]$

given $E[z]$ [where “ $E[\log L]$ ” is wrt random $z \sim E[z] = p(z=1)$]

Bayes

MLE

EM Issues

Under mild assumptions, EM is guaranteed to increase likelihood with every E-M iteration, hence will *converge*.

But it may converge to a *local*, not global, max.
(Recall the 4-bump surface...)

Issue is intrinsic (probably), since EM is often applied to problems (including clustering, above) that are *NP-hard* (so fast alg is unlikely)

Nevertheless, widely used, often effective

Applications

Clustering is a remarkably successful exploratory data analysis tool

- Web-search, information retrieval, gene-expression, ...

- Model-based approach above is one of the leading ways to do it

Gaussian mixture models widely used

- With many components, empirically match arbitrary distribution

- Often well-justified, due to “hidden parameters” driving the visible data

EM is extremely widely used for “hidden-data” problems

- Hidden Markov Models – speech recognition, DNA analysis, ...

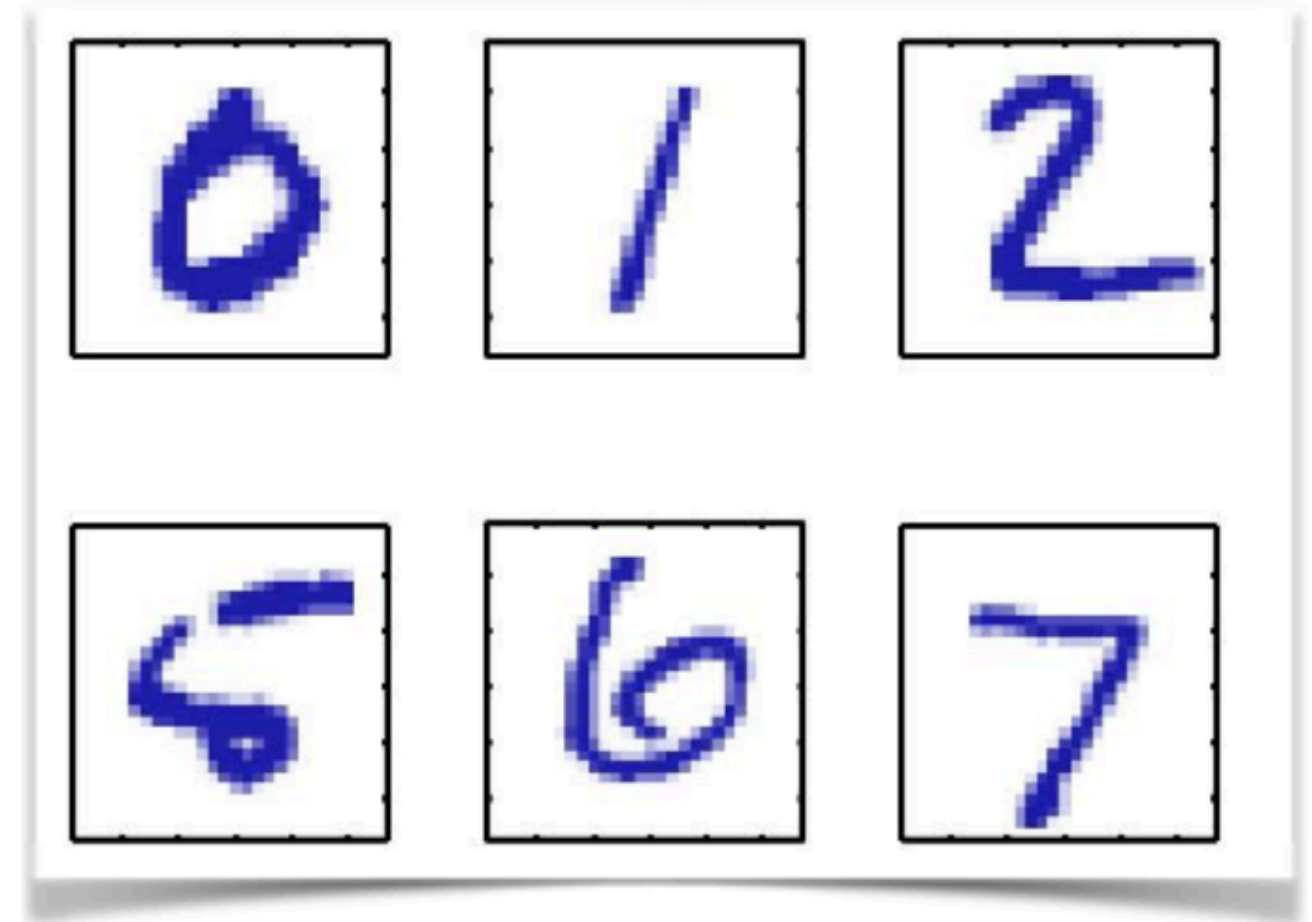
A “Machine Learning” Example

Handwritten Digit Recognition

Given: 10^4 unlabeled, scanned images of handwritten digits, say 25 x 25 pixels,

Goal: automatically classify new examples

Possible Method:



Each image is a point in $\mathbb{R}^{625}$; the “ideal” 7, say, is one such point; model other 7’s as a Gaussian cloud around it

Do EM, as above, but 10 components in 625 dimensions instead of 2 components in 1 dimension

“Recognize” a new digit by best fit to those 10 models, i.e., basically max E-step probability